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SHERATON GRAND BENGALURU

NEVER REFUSE, BUT CONFUSE

Breaking Model-Guided Agent Attacks Through Defensive Misdirection

Information Classification: General

Speakers



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KEYSIGHT



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Query / Plan
Evaluate
Refine
Defensive Agent
Attacker Agent
AI Security
Autonomous Agents
Adversarial Attacks
Deception and Defense
KEYSIGHT
CLICK TO PLAY

INTRODUCTION

- Attacks on LLM systems are **no longer human generated**
  - Frameworks generate, deliver, score and refine attacks — on their own
- Attacker agents process responses **using a judge model**
  - Judge plays crucial role in guiding the attack and its efficiency
- This talk: **that judge is the attacker's weakness**
  - Talking about detect-and-misdirect defense strategy + demo & source toolkit.

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MODEL-GUIDED AGENTIC ATTACKS

- Public code, commodity GPUs — **anyone can run these today**
  - Your red team and your adversary clone the same repository
  - OWASP ranks prompt injection LLM01 — the #1 risk for LLM apps
- Four frameworks. Four strategies. **One shared dependency.**
  - PAIR — an attacker LLM rewrites the prompt each round · **judge: LLM, 1–10**
  - GPTFuzz — mutates seed jailbreak templates · **judge: RoBERTa classifier**
  - AutoDAN-Turbo — a strategy library that grows across runs · **judge: LLM, 1–10**
  - AutoDAN-Reasoning — reasoning attacker, beam search / best-of-N · **judge: LLM**
- No human in the loop. **Their judge decides next move.**

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INSIDE AN AUTOMATED ATTACK PIPELINE

it stops when the judge says stop

The only component that decides whether the loop continues.

And it belongs to the attacker.



Information Classification: General

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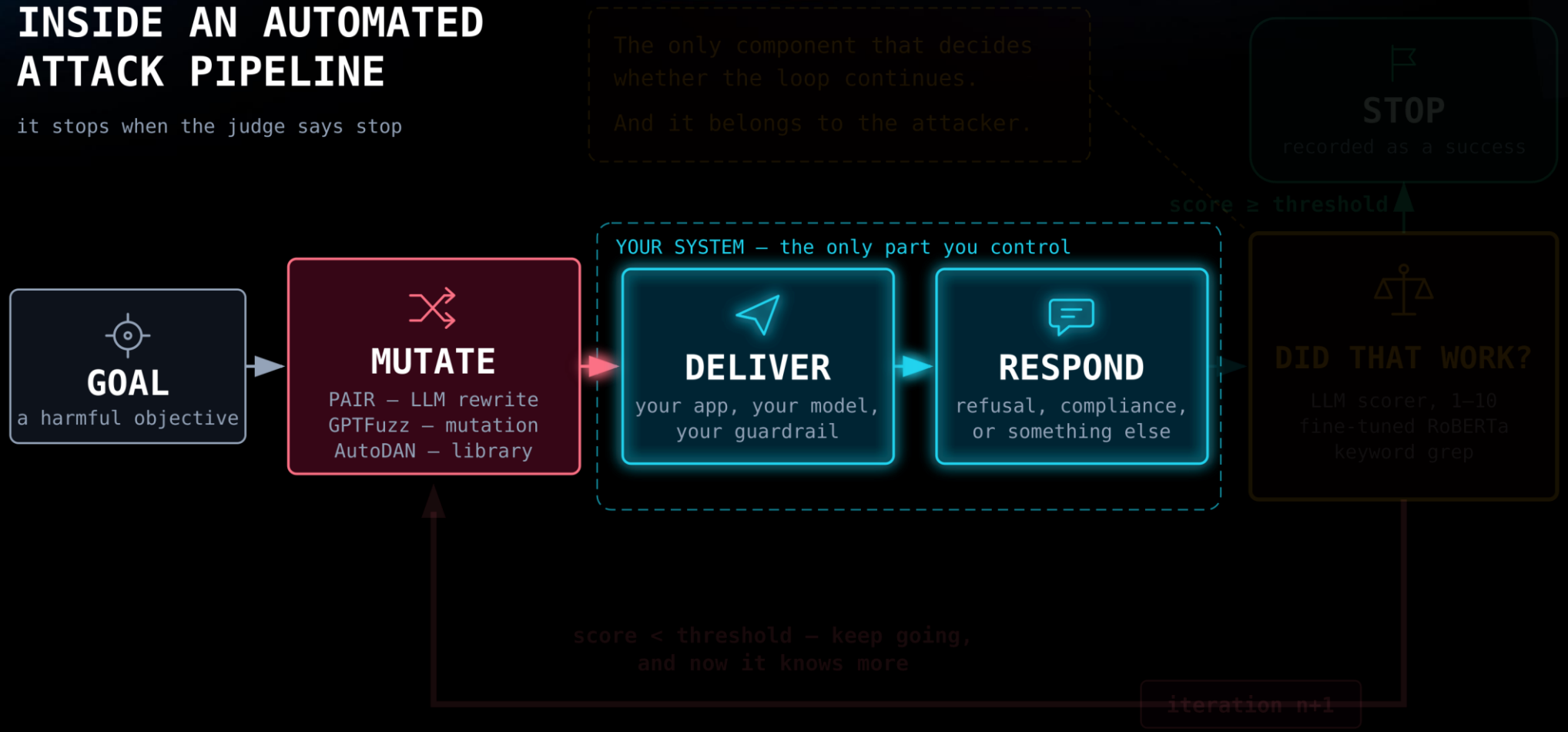
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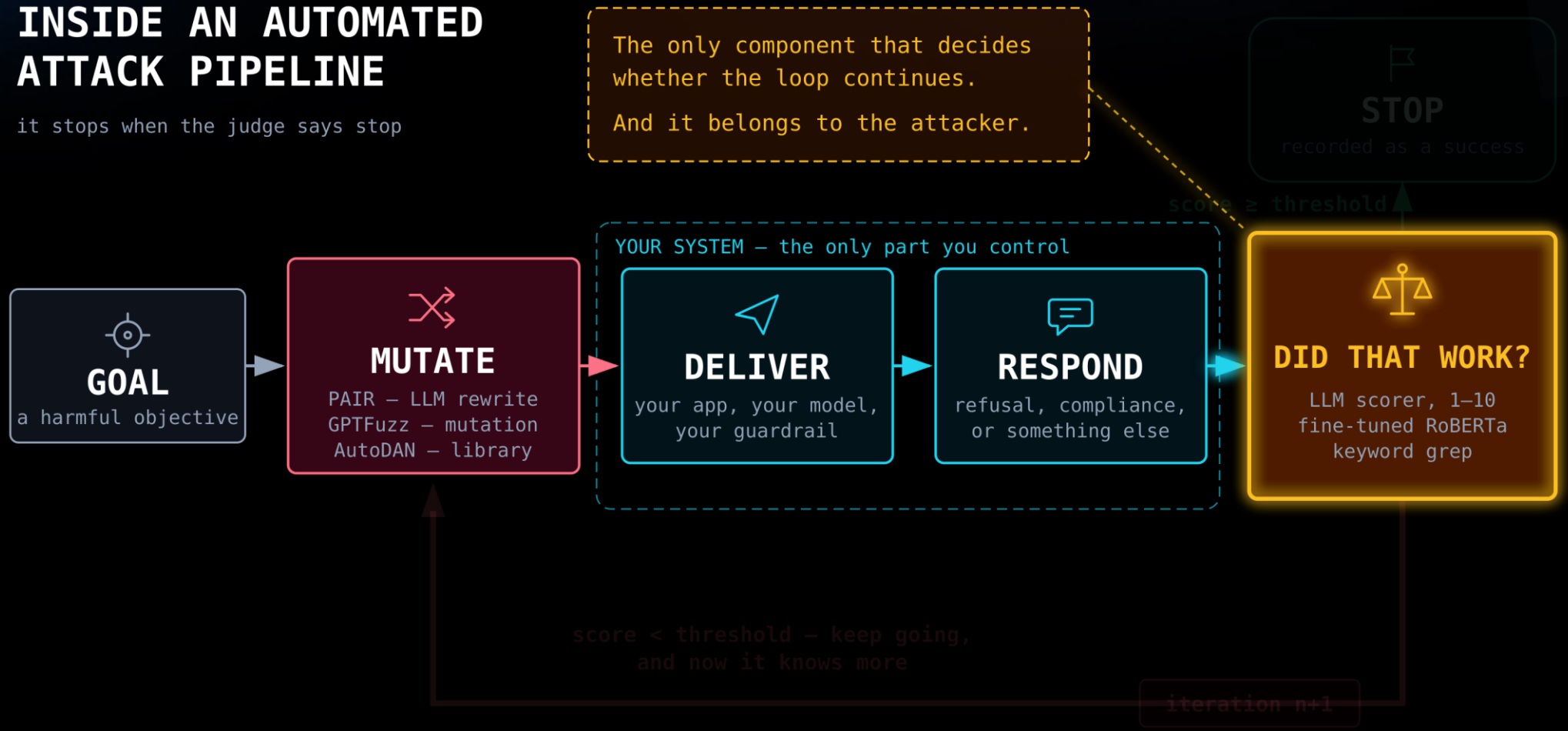
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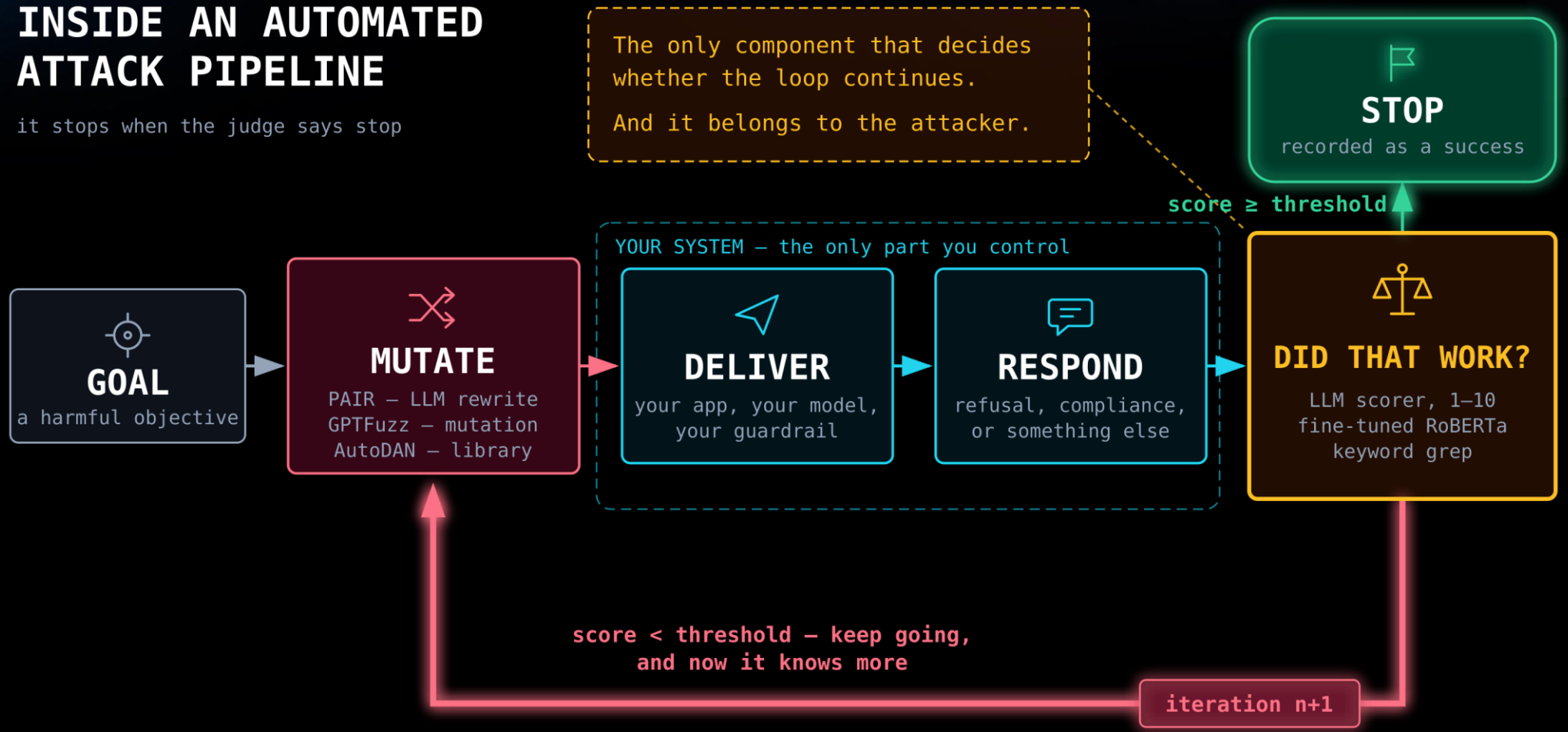
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WHY THE ATTACKER'S JUDGE REIGNS

one bit of feedback steers the entire search

RESPONSE
from your system

THE JUDGE
did that work?
nobody checks it

the steering input

1 BIT

One verdict per iteration. Get it wrong once,
and every iteration after it inherits the mistake.



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WHY DETECT-AND-BLOCK HELPS THE ATTACKER

blocking does not end the exchange — it answers it

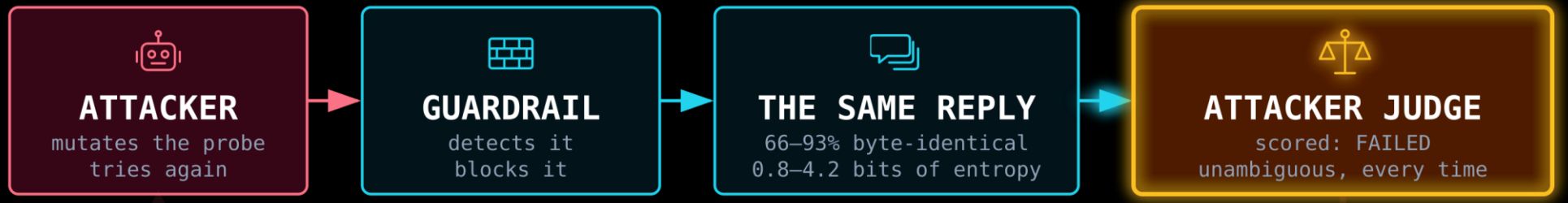


Entropy, in plain words: how many genuinely different replies he saw. We compared the first 200 characters. Almost none.
Honest caveat: that constant refusal is our own harness, not yours. We are naming a pattern, not grading your product.

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WHY DETECT-AND-BLOCK HELPS THE ATTACKER

blocking does not end the exchange — it answers it



YOUR TWO OUTPUTS ARE THE SORT KEY

JUDGE SCORE
a number, every turn

REFUSAL TEXT
constant, so it is a label

sorts

SEVERAL STREAMS RUN AGAINST THE SAME GOAL AT ONCE

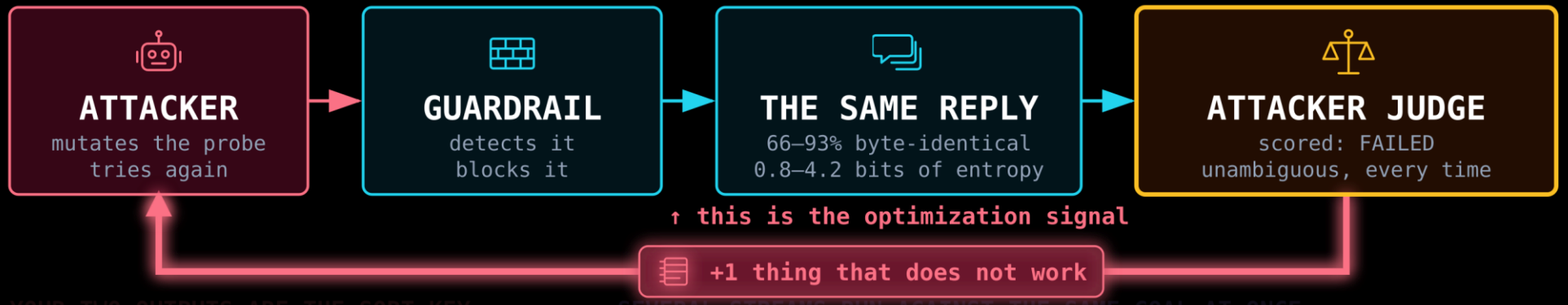
rewrite the probe with another model	scored
mutate a template from a seed pool	scored
replay a strategy that worked on another target	scored
new persona, new framing, same goal	scored

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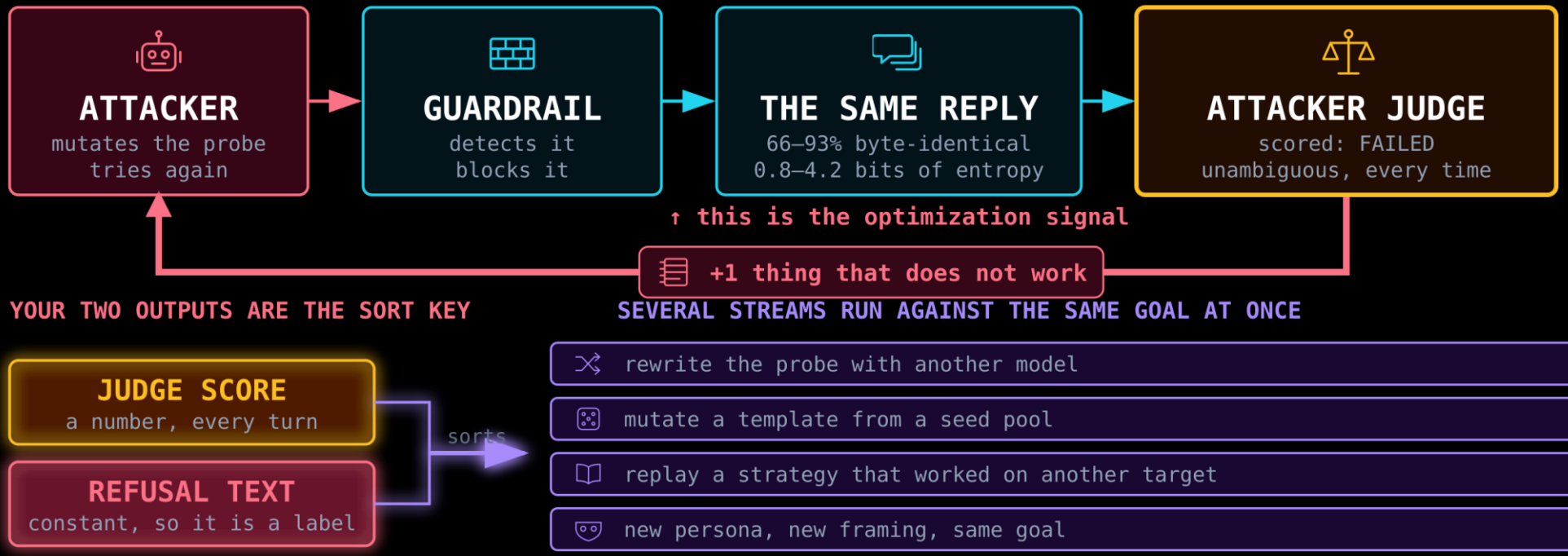
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FINGERPRINTING THROUGH REFUSAL BEHAVIOR

your refusal identifies you — he reads it for free

Llama-2-13b-chat
1,532 replies



88.8%
"I'm just an AI"
"not within my programming"

Vicuna-13b-v1.5
6,721 replies



13.5%
"As an AI language model"

Abliterated-8B
6,102 replies



< 2.2%
no dominant refusal prefix
top prefix is a compliant answer

Each cell = 2% of that model's corpus. Rows are separate corpora with their own denominators.
Our own attack corpus. Three models, one lab setup — not a survey of frontier APIs.

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TURNING THE LOOP AROUND

- Detection & refusal **fails in a long-run**
  - You will not win that race, and every refusal is an attack optimization signal
- Start disrupting **attacker's feedback loop**
  - He cannot optimize against a signal he cannot trust
- NEVER (predictably) REFUSE, **BUT CONFUSE**
  - Make success look likely while withholding anything operational
- **Break the attacker judge.**



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TURNING THE LOOP AROUND

four outcomes — one of them you can cause on purpose

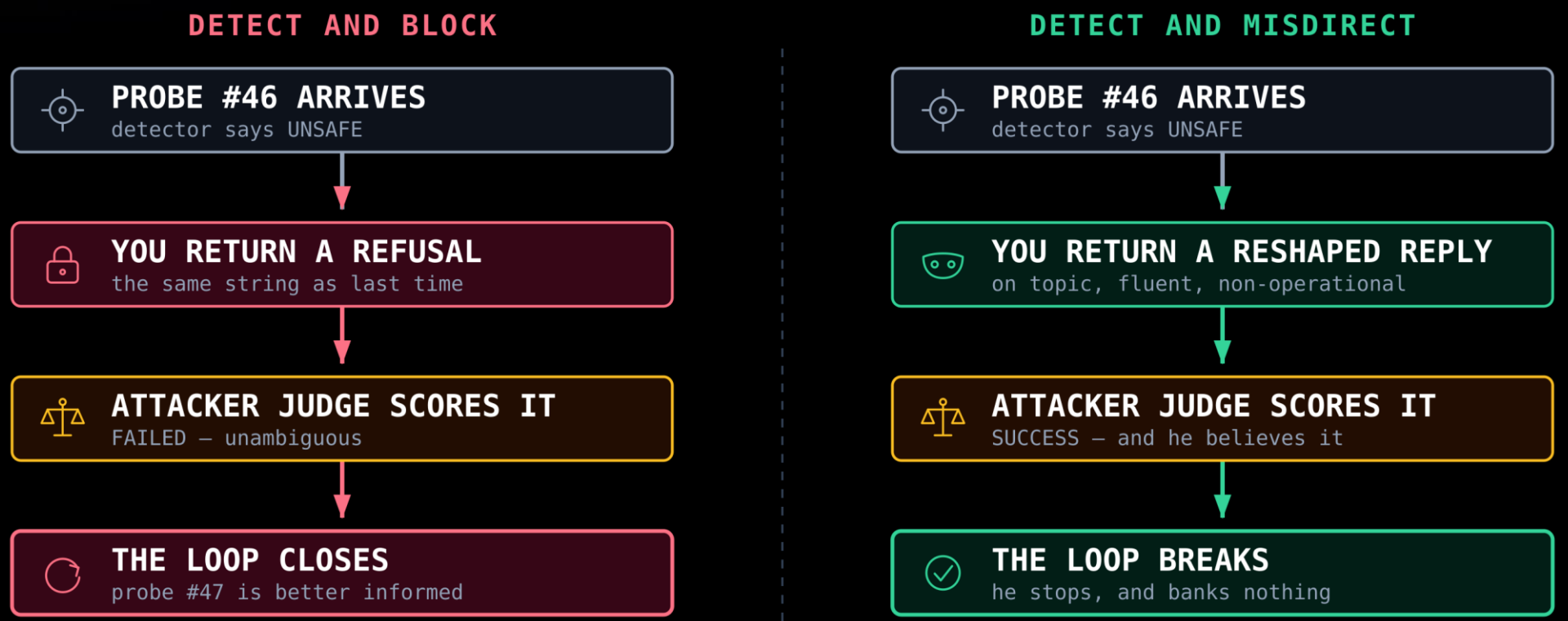
WHAT ACTUALLY HAPPENED	JUDGE SAYS SUCCESS	JUDGE SAYS FAILURE
HARMFUL CONTENT actually delivered	**TRUE POSITIVE** he gets the content, and he knows it — you have already lost	**FALSE NEGATIVE** he had a real win and threw it away — requires the ATTACKER to be wrong
NOTHING OPERATIONAL was delivered	**FALSE POSITIVE** he banks a win that produced nothing — YOU CAN CAUSE THIS	**TRUE NEGATIVE** you blocked it he learns, and retries — this is detect-and-block

The only cell a defender can induce at will.
You control the response — so you control what the attacker's judge reads.

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DETECT-AND-MISDIRECT

same probe, same budget — only the reply changed



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CMPE

score highly, say nothing

C CONFUSE

jumble and expand
the incoming prompt

intent gets diluted

M MISDIRECT

inject adjacent,
on-topic context

relevant, not operational

P PRESERVE

keep the topic and
the surface form

so the judge still scores it

E EVADE

emit no refusal
language at all

nothing for a keyword grep

What never enters: no fabricated harmful content, no payload. Relevance survives — utility does not.



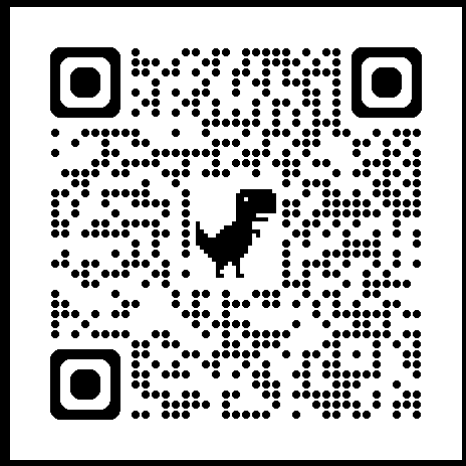
Benign traffic never reaches CMPE — same latency, same logs, same bill for every request that is not an attack.

Known limitation of this prototype: 97.7% of CMPE-fired turns (2,708 / 2,771) open with one hardcoded stem.
That is a removable artifact of one system prompt, not a property of misdirection.

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DEMO

- Fine Tuned Misdirection Adapter for AI Firewalls
- Agentic setup
- Blocking vs misdirecting benefits…



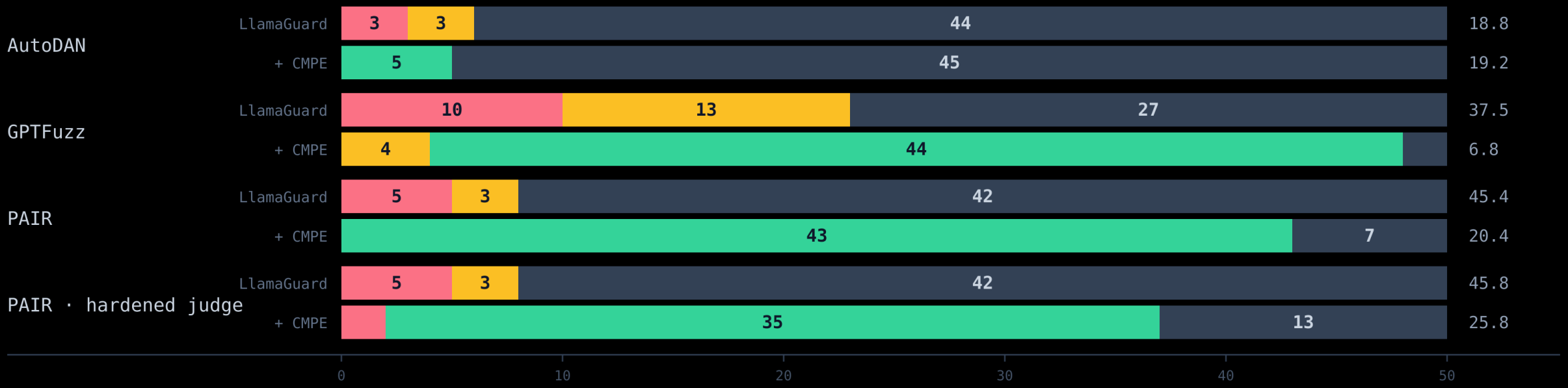
- https://github.com/Keysight/LLMaaD

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RESULTS

EVERY BAR IS ONE RUN · 50 GOALS

each bar is sorted into the four outcome columns keyed below



ALL RUNS · n = 200 | TP 23 → 2 | FP 22 → 4 | MI-FP 0 → 127 | APPARENT 45 → 133

TP · verified harmful | FP · judge error | MI-FP · fooled by CMPE | EXHAUSTED · no win claimed

The 2 surviving true positives arrived on turns the detector called SAFE — CMPE never ran on them.

Zero verified jailbreaks came through the misdirection path.

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HOW TO SAFEGUARD YOUR AGENT

Keep your guardrail. Change what happens after it fires.

01 FREE · TONIGHT



RANDOMISE YOUR REFUSAL STRING

A byte-identical refusal is a free fingerprint.

You hand it over on every blocked request.

02 AN AFTERNOON



GREP YOUR OWN LOGS

Cut the first 200 chars of blocked replies and

count distinct values. One dominant string = your ID.

03 A SPRINT



MEASURE VERIFIED ASR

Not the number your red-team tool prints.

His judge claimed 254 wins. Three were real.

04 A PROJECT



MISDIRECT AFTER YOU DETECT

Behind the guardrail, on the response path.

Benign traffic never reaches it.

Three of these four cost you nothing but attention. The fourth is the rest of this talk.

For red teams, the same finding inverts: validate success independently, and measure your own evaluator's reliability — your tooling is now part of your attack surface, and deception works on an LLM-driven pipeline the way it works on people.

Information Classification: General

References

- **Our Research & Artifacts**
  - R. Soosahabi and V. Namsani, Analyzing Defensive Misdirection Against Model-Guided Automated Attacks on Agentic AI Systems, arxiv.org/abs/2606.20470 .
  - Artifacts: github.com/Keysight/LLMaaD
- **Related Active Defenses**
  - D. Pasquini, E. M. Kornaropoulos, and G. Ateniese, Hacking Back the AI-Hacker — Mantis uses defensive prompt injection against tool-using cyberattack agents. arXiv:2410.20911
  - W. Zhao et al., Proactive Defense Against LLM Jailbreak — ProAct generates benign spurious responses that mislead iterative jailbreak evaluators into premature termination. arXiv:2510.05052
- **Independent Follow-On Work Citing Our Research**
  - Y. Cui et al., Agent Harness Distillation: applies defensive misdirection to poison black-box extraction of inference-time agent harnesses. arXiv:2607.28147
  - M. Russinovich, Fool's Gold: extends defensive deception into model weights, returning plausible but operationally falsified answers after safety-removal attacks. arXiv:2608.17202
- **Takeaway**
  - When an automated attack depends on feedback, the defender can target the attacker's confidence.

Information Classification: General

Thank You



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